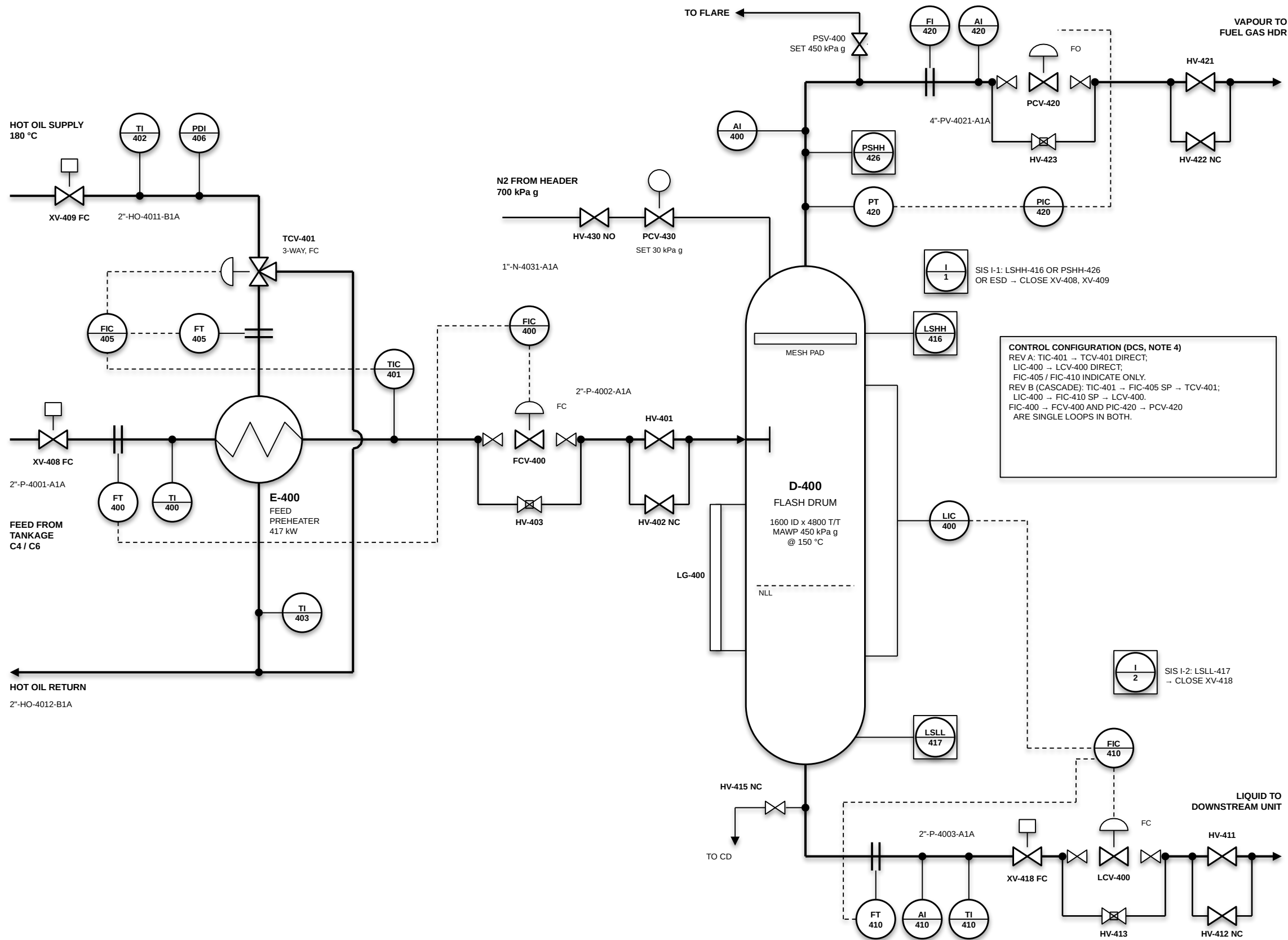




**GENERAL NOTES**

- DESIGN BASIS FROM THE LEGACY 309-TYPE TRAINING BOOKLET (METRIC): FEED 16 m<sup>3</sup>/h AT 10 °C, 35 wt% n-C4 / 65 wt% n-C6; TIC-401 74 °C; PIC-420 102.7 kPa g; LIC-400 50 %. OTHER VALUES CALCULATED (IDEAL RAOULT VLE, ADIABATIC FLASH): VAPOUR 1.72 t/h (= 610 Nm<sup>3</sup>/h), 76.8 wt% C4; LIQUID 8.49 t/h (14.0 m<sup>3</sup>/h), 26.4 wt% C4, 49 °C; E-400 DUTY 417 kW.
- ASSUMED, TBC: FEED SUPPLY 600 kPa g; HOT OIL 180 °C, 150 kPa HEADER ΔP; VAPOUR HEADER 30 kPa g; DOWNSTREAM 40 kPa g; D-400 SIZE AND MAWP; LINE SIZES AND SPECS PRELIMINARY.
- PSV-400 RELIEF CASES (BLOCKED VAPOUR OUTLET, FIRE, TCV-401 FAILED OPEN) AND SIZE: HOLD FOR RELIEF STUDY.
- REV B CASCADE IS A DCS CONFIGURATION ON THE SAME FIELD INSTRUMENTS; THE OPERATOR SELECTS REV A OR REV B.
- CONTROL-VALVE STATION ISOLATION VALVES LOCKED OPEN IN OPERATION; DRAINS AND VENTS ON EACH STATION NOT SHOWN.
- PRESSURE INSTRUMENTS c/w ROOT AND TEST VALVES (TYP., NOT SHOWN).

**SAFETY AND OPERATING NOTES**

- SIS I-1: LSHH-416 (90 %, 5 s) OR PSHH-426 (250 kPa g, 3 s) OR ESD → CLOSE XV-408 FEED AND XV-409 HOT OIL. LATCHED.
- SIS I-2: LSL-417 (5 %, 5 s) → CLOSE XV-418 (GAS BLOW-BY TO THE DOWNSTREAM UNIT). OPEN PERMITTED ABOVE 10 %.
- LSHH-416 / LSL-417 / PSHH-426 ARE INDEPENDENT OF LT-400 / PT-420 (SEPARATE SENSORS AND TAPS).
- PCV-430 N2 BLANKET HOLDS D-400 ≥ 30 kPa g (VACUUM PROTECTION ON LOSS OF HEAT); AI-400 = N2 IN DRUM VAPOUR.
- HV-401/402, HV-411/412, HV-421/422: LEGACY BLOCK VALVE WITH PARALLEL BYPASS BLOCK VALVE (NORMALLY CLOSED).
- FAIL-SAFE: FCV-400 FC, LCV-400 FC, TCV-401 FC (FULL BYPASS, NO HEAT), PCV-420 FO (DRUM TO HEADER).
- ALARMS: FIC-400 5/18 m<sup>3</sup>/h; TIC-401 69/79 °C; LIC-400 20/80 %; PIC-420 70/130 kPa g; AI-410 H 28 wt%; AI-420 L 75 wt%; AI-400 H 5 mol%. CONTROLLER TUNING: SEE DWG FD-400-PID-002.

**LEGEND**

— PROCESS LINE  
— UTILITY / SECONDARY LINE  
- - - ELECTRONIC SIGNAL

⊗ GATE / BLOCK VALVE (MANUAL)  
⊗ GLOBE BYPASS VALVE (MANUAL, THROTTLING)  
⊗ CONTROL VALVE, DIAPHRAGM ACTUATOR  
⊗ ON/OFF SHUT-OFF VALVE (SIS)  
⊗ SELF-ACTING PRESSURE REGULATOR  
⊗ ORIFICE FLOW ELEMENT  
⊗ PRESSURE SAFETY VALVE

XX nnn FIELD-MOUNTED INSTRUMENT  
XX nnn DCS FUNCTION (OPERATOR ACCESSIBLE)  
XX nnn SIS FUNCTION / INTERLOCK (SEPARATE SENSOR)

FC / FO FAIL CLOSED / FAIL OPEN ON AIR OR SIGNAL LOSS  
NO / NC NORMALLY OPEN / NORMALLY CLOSED  
NLL NORMAL LIQUID LEVEL  
CD CLOSED DRAIN

**LINE NUMBER: SIZE - SERVICE - NUMBER - SPEC**

P PROCESS LIQUID PV PROCESS VAPOUR HO HOT OIL  
N NITROGEN FL FLARE CD CLOSED DRAIN

**EQUIPMENT**

D-400 FLASH DRUM, VERTICAL, 1600 ID x 4800 T/T  
E-400 FEED PREHEATER, SHELL & TUBE, HOT OIL  
SHELL / FEED TUBES, DUTY 417 KW

| REV                                        | DESCRIPTION                                | DATE            |                 |
|--------------------------------------------|--------------------------------------------|-----------------|-----------------|
| 0                                          | ISSUED FOR TRAINING (OTS FLASH DRUM VER 1) | 24-SEP-26       |                 |
| CHEM. ENG., CHULALONGKORN UNIVERSITY — OTS |                                            |                 |                 |
| FLASH DRUM D-400 — P&ID                    |                                            |                 |                 |
| DRAWN:<br>A. SOOTTITANTAWAT                | CHK'D: --                                  | APP'D: --       | DATE: 24-SEP-26 |
| DWG NO.<br>FD-400-PID-001                  | REV<br>0                                   | SCALE<br>N.T.S. | SHEET<br>1 OF 2 |